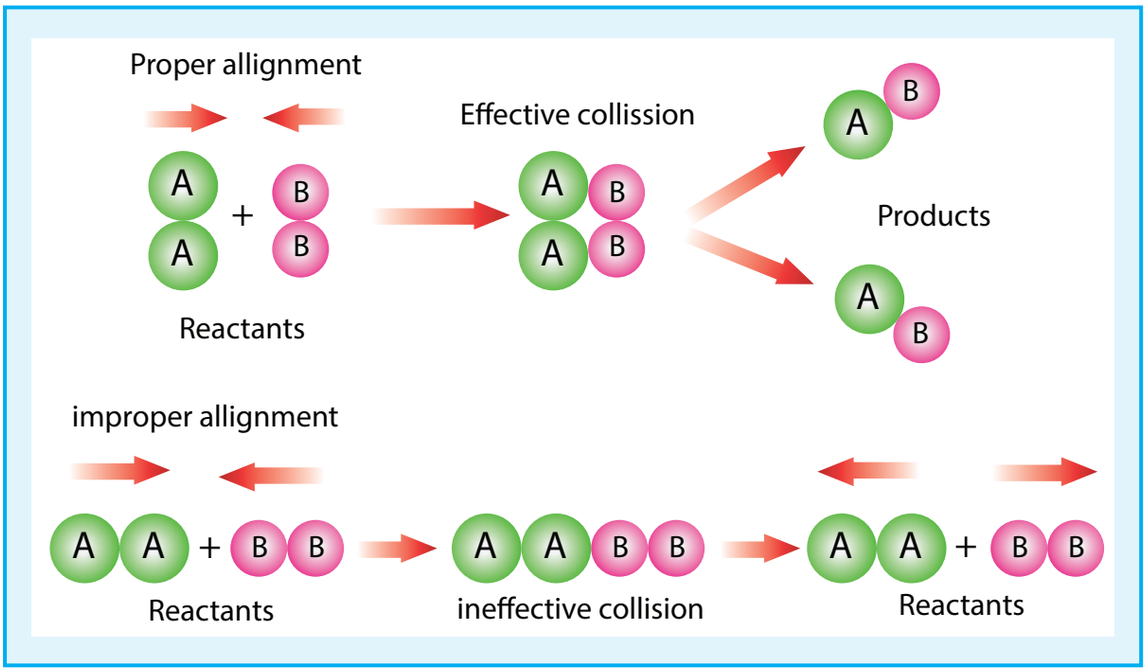


Fig 7.6 - Orientation of reactants - schematic representation

orientation factor i.e., even if the reactant collide with sufficient energy, they will not react unless the orientation of the reactant molecules is suitable for the formation of the transition state.

The figure 7. 6 illustrates the importance of proper alignment of molecules which leads to reaction.

The fraction of effective collisions (f) having proper orientation is given by the steric factor p.

$\Rightarrow$ Rate = p x f x collision rate

i.e., $\text{Rate} = p \times e^{\frac{-E_a}{RT}} \times Z\,[A_2][B_2]$ ...(1)

As per the rate law,

$\text{Rate} = k\,[A_2]\,[B_2]$ ...(2)

Where k is the rate constant

On comparing equation (1) and (2), the rate constant k is '

$k = p\,Z\,e^{\frac{-Ea}{RT}}$

7.8 Arrhenius equation – The effect of temperature on reaction rate

Generally, the rate of a reaction increase with increasing temperature. However, there are very few exceptions. The magnitude of this increase in rate is different for different reactions. As a rough rule, for many reactions near room temperature, reaction rate tends to double when the temperature is increased by 10^0C .

Activity

Let us understand the effect of temperature on reaction rate by doing this activity.

i. Take two test tubes, label them as A and B

ii. Take 5 ml of cold water in A, add a drop of phenolphthalein indicator and then add Magnesium granules.

iii. Repeat the above with 5 ml of hot water in test tube B.

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